

Abhinav Atul

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Summary

CS undergraduate at VIT Bhopal (CGPA: 8.71) and IIT Madras BS Data Science, with experience building distributed systems, agentic AI pipelines, and real-time ML applications. Proficient in C++, Python, and Java with strong foundations in DSA, system design, and scalable backend development.

Technical Skills

Languages: C++ (Primary), Java, Python, SQL, JavaScript, TypeScript

Systems: Data Structures & Algorithms, Operating Systems, Distributed Systems, Concurrency, Multi-threading

Core CS: DBMS, Computer Networks, OOP (Encapsulation, Inheritance, Polymorphism, Abstraction)

Frameworks & Tools: FastAPI, React, PostgreSQL, Supabase, OpenCV, Git, YOLOv8, Ollama

AI & ML: On-device LLM Inference (Ollama), Supervised Learning, CNN, Feature Engineering, scikit-learn

Education

VIT Bhopal University

Bhopal, India

B.Tech in Computer Science and Engineering — Expected May 2028

Aug 2024 – Present

– CGPA: 8.71 / 10 | Coursework: DSA, OS, DBMS, Computer Networks, OOP, Probability & Statistics

Projects

RegisterSync AI — Regulatory Compliance Automation

GitHub

Agentic AI System | Hackathon Project

Python, FastAPI, React, Ollama, PyMuPDF, PostgreSQL

- * Built a multi-agent AI platform ingesting RBI/SEBI PDFs, extracting compliance rules via locally hosted Ollama LLM, and auto-routing tasks to departments — no cloud API dependency.
- * Designed conflict-detection and verification agents; exposed 7+ FastAPI endpoints with PyMuPDF chunking and on-device deepseek-r1 for privacy-preserving inference.

Smart Transit System

GitHub

Distributed Backend + Real-Time ML System

Python, FastAPI, PostgreSQL, YOLOv8

- * Built a distributed real-time transit platform with vehicle tracking, ETA prediction, and passenger analytics across concurrent data streams using 10+ RESTful APIs.
- * Integrated YOLOv8-based crowd detection for real-time passenger flow monitoring at production-grade inference speeds.

IoT Intrusion Detection System

GitHub

ML-Based Network Security System

Python, scikit-learn, Feature Engineering

- * Built an end-to-end ML pipeline on CICIDS2017 (2.8M+ records, 78 features) classifying traffic across 7 attack categories; evaluated Random Forest and KNN with feature selection to handle class imbalance.

Driver Drowsiness Detection System

GitHub

Real-Time Computer Vision System

Python, OpenCV, CNN

- * Developed a real-time fatigue detection pipeline using CNN and facial landmark analysis, achieving **84.9% accuracy** with sub-second alert response on live video feeds.

Experience & Activities

LINPACK Club — Technical Team Member

Aug 2025 – Present

- * Contributing to internal web platform development using React and Supabase in a cross-functional team of 10+ members.

GirlScript Summer of Code (GSSoC) 2026 — Open Source Contributor

- * Ranked in the **top 3%** among 43,586 contributors; contributing features and bug fixes to production JavaScript/React codebases across multiple repositories.

Smart India Hackathon 2025

- * Led backend development for a real-time smart transit analytics system in a 6-member team, integrating distributed data pipelines within a 36-hour build cycle.

Certifications

Google AI Essentials – Google | Oracle Cloud Infrastructure 2025 Certified Data Science Professional | The Bits and Bytes of Computer Networking – Google